Example 5.8 (Illustrating SAS Similarity)

If A is the midpoint of *RU* and T is the midpoint of *RN*, prove that $\Delta RAT \sim \Delta RUN$.

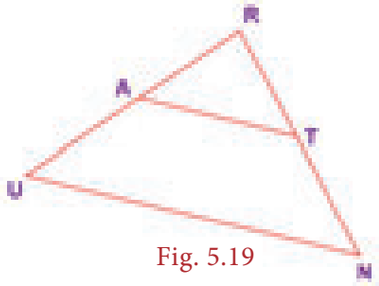


Fig. 5.19

Proof:

	Statements	Reasons
1	$\angle ART = \angle URN$	$\angle R$ is common in ΔRAT and ΔRUN
2	$RA = AU = \frac{1}{2} RU$	A is the midpoint of *RU*
3	$RT = TN = \frac{1}{2} RN$	T is the midpoint of *RN*
4	$\frac{RA}{RU} = \frac{RT}{RN} = \frac{1}{2}$	the sides are proportional from 2 and 3
5	$\Delta RAT \sim \Delta RUN$	by SAS (1 and 4)

Example 5.9 (Illustrating SSS similarity)

Prove that $\Delta PQR \sim \Delta PRS$ in the given Fig. 5.20.

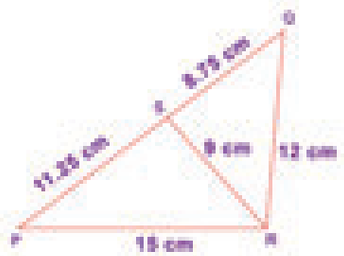


Fig. 5.20

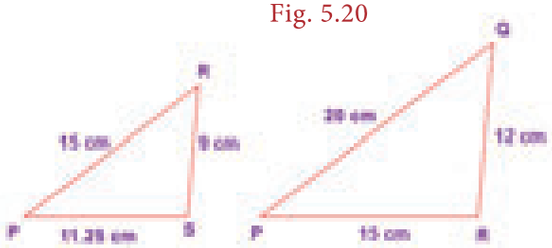
Solution:

Now, $\frac{PQ}{PR} = \frac{20}{15} = \frac{4}{3}$

$\frac{PR}{PS} = \frac{15}{11.25} = \frac{4}{3}$

Also, $\frac{QR}{RS} = \frac{12}{9} = \frac{4}{3}$

We find $\frac{PQ}{PR} = \frac{PR}{PS} = \frac{QR}{RS}$



That is, their corresponding sides are proportional.

$\therefore$ By SSS Similarity, $\Delta PQR \sim \Delta PRS$

Example 5.10 (Illustrating RHS similarity)

The height of a man and his shadow form a triangle similar to that formed by a nearby tree and its shadow. What is the height of the tree?

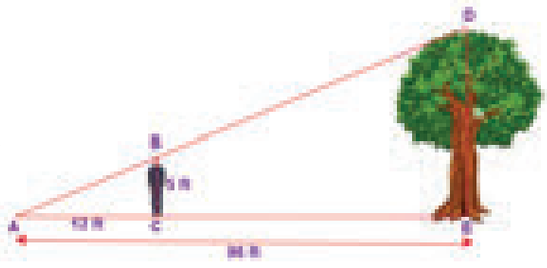


Fig. 5.21

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